

Direct Generation of a Somos–4 Sequence from an Algebraic Generating Function

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Abstract

We record a five-parameter quadratic algebraic generating function whose coefficient sequence has prescribed initial Hankel determinants (λ, m, r, η) and whose Hankel transform belongs to the Somos–4 family $A(1, \tau)$. The Hankel determinants and their parameters are introduced first. We then expose the underlying Stieltjes continued-fraction construction, including its normalization and coefficient recurrence, before presenting the explicit algebraic equation. We subsequently describe the two square-root branches, their coalescence at the origin, a desingularized coefficient recurrence, the induced shift of the Hankel parameters, and the nonuniqueness of the underlying moment representation. We make explicit that the construction fixes this nonuniqueness by prescribing, in addition to the ordinary Hankel transform, a companion Hankel sequence equal to the same Somos–4 orbit with its first entry removed.

1 Context and relation to earlier work

The relation between Hankel determinants, continued fractions, moment sequences, and nonlinear recurrences has a long history. Classical continued-fraction treatments include Wall [19] and Jones and Thron [12]; the combinatorial theory of continued fractions was developed systematically by Flajolet [5]. Determinantal evaluations and transformations relevant to Hankel matrices are discussed, for example, by Krattenthaler [13], Layman [14], and French [7]. The quadratic transformation of Sulanke and Xin [17] and related continued-fraction methods of Gessel and Xin [9] have become particularly effective tools for proving structured Hankel-determinant identities.

Somos recurrences originated in the investigations of Somos and were popularized by Gale [8]. Their Laurent phenomenon and their connections with elliptic divisibility sequences and elliptic curves are part of a broader theory; see, among others, Fomin and Zelevinsky [6], van der Poorten and Swart [18], Hone [10], and Hone and Swart [11]. Xin proved the original Somos–4 Hankel determinant conjecture by a continued-fraction recursion system [21].

A substantial sequence of papers by Barry studies generating functions whose Hankel transforms satisfy Somos-type recurrences. These works include generalized Catalan recurrences [1], generalized Narayana triangles [2], Riordan–Bernstein constructions [3], and Riordan pseudo-involutions, continued fractions, and elliptic curves [4]. Barry’s examples and conjectures form an important part of the immediate background of the present construction.

Wang and Zhang subsequently formulated a sufficient condition for an algebraic generating function to have an (α, β) Somos–4 Hankel transform, using the Sulanke–Xin machinery [20]. Liu, Wang, and Zhang extended this condition over a rational-function field and used it to settle the outstanding Barry conjectures treated there; they also studied odd and even Somos subsequences, the Hurwitz transform, orthogonal-polynomial Hankel formulas, periodic Hankel determinants, and elliptic parameter families [15].

The direction taken here is complementary. Rather than beginning with a given generating function and deriving the parameters of its Hankel recurrence, we begin with prescribed initial Hankel data (λ, m, r, η) and the Somos coefficient τ , impose a companion Hankel orbit, and

construct an explicit quadratic algebraic generating function realizing those data. The explicit companion prescription is also what selects one moment representation among the generally nonunique representations having the same ordinary Hankel transform.

2 Hankel determinants and Somos parameters

Let

$$G(z) = \sum_{n \geq 0} g_n z^n$$

be a formal power series, and for $n \geq 1$ define its ordinary Hankel determinants by

$$H_n = \det(g_{i+j})_{0 \leq i, j \leq n-1}, \quad H_0 = 1. \quad (1)$$

The five parameters $\lambda, m, r, \eta, \tau$ have the following roles:

$$H_1 = \lambda, \quad H_2 = m, \quad H_3 = r, \quad H_4 = \eta, \quad (2)$$

and the Hankel sequence satisfies the Somos-4 recurrence

$$H_{n+4}H_n = H_{n+3}H_{n+1} + \tau H_{n+2}^2. \quad (3)$$

Thus τ is the second coefficient of the Somos family $A(1, \tau)$, whereas λ, m, r , and η select the initial Hankel determinants. The algebraic generating function producing these determinants is given in the next section. The construction makes one additional choice that is easy to overlook: together with $(H_n)_{n \geq 0}$, it fixes a companion Hankel sequence

$$\hat{H}_n = H_{n+1} \quad (n \geq 0). \quad (4)$$

Thus the companion sequence is the same Somos-4 orbit with its first entry removed. This second prescription is part of the definition of the selected moment representation; it is not a consequence of the ordinary Hankel determinants alone. Section 8 explains why prescribing both sequences removes the degree of freedom discussed there.

3 Behind the construction: the Stieltjes fraction

The quadratic equation below may be obtained by first constructing a Stieltjes continued fraction whose Hankel determinants follow the prescribed Somos-4 orbit. Write

$$G(x) = \frac{c_0}{1 - \frac{c_1 x}{1 - \frac{c_2 x}{1 - \frac{c_3 x}{\ddots}}}}. \quad (5)$$

For this normalization, the ordinary Hankel determinants are [19, 12, 5]

$$H_n = c_0^n \prod_{j=1}^{n-1} (c_{2j-1} c_{2j})^{n-j}. \quad (6)$$

Thus

$$H_1 = c_0, \quad H_2 = c_0^2 c_1 c_2, \quad H_3 = c_0^3 c_1^2 c_2^2 c_3 c_4,$$

and similarly at higher orders.

It is useful first to remove the exponential scaling caused by H_1 . Set

$$a_1 = 1, \quad a_2 = \frac{m}{\lambda^2}, \quad a_3 = \frac{r}{\lambda^3}, \quad a_4 = \frac{\eta}{\lambda^4}. \quad (7)$$

Then $H_n = \lambda^n a_n$ for $1 \leq n \leq 4$, and the normalized sequence (a_n) obeys the same Somos-4 recurrence,

$$a_{n+4}a_n = a_{n+3}a_{n+1} + \tau a_{n+2}^2. \quad (8)$$

The powers of λ in (7) are essential: for nonunit $H_1 = \lambda$, the correct normalization is $a_j = H_j/\lambda^j$, not H_j/λ^{j-1} .

A convenient initial segment of the Stieltjes coefficients is

$$c_0 = \lambda, \quad c_1 = a_3, \quad c_2 = \frac{a_2}{a_3}, \quad (9)$$

$$c_3 = \frac{a_4}{a_2 a_3}, \quad c_4 = \frac{a_3^2}{a_2 a_4}, \quad c_5 = \tau \frac{a_2 a_3}{a_4} + \frac{a_2^2}{a_3}. \quad (10)$$

Equivalently, in the parameters used throughout this paper,

$$\begin{aligned} c_0 &= \lambda, & c_1 &= \frac{r}{\lambda^3}, & c_2 &= \frac{\lambda m}{r}, \\ c_3 &= \frac{\lambda \eta}{mr}, & c_4 &= \frac{r^2}{m\eta}, & c_5 &= \frac{\tau mr}{\lambda \eta} + \frac{m^2}{\lambda r}. \end{aligned} \quad (11)$$

The remaining coefficients are generated alternately by

$$c_{2k} = \frac{1}{c_{2k-2}c_{2k-1}}, \quad c_{2k+1} = c_{2k-2} + \frac{\tau}{c_{2k-1}} \quad (k \geq 3). \quad (12)$$

In particular,

$$c_6 = \frac{1}{c_4 c_5}, \quad c_7 = c_4 + \frac{\tau}{c_5}, \quad c_8 = \frac{1}{c_6 c_7}, \quad c_9 = c_6 + \frac{\tau}{c_7}.$$

Substitution in (6) gives $H_j = \lambda^j a_j$ for $1 \leq j \leq 4$. At the next order it gives

$$\frac{H_5}{\lambda^5} = \frac{a_3 a_4 c_5}{a_2} = a_2 a_4 + \tau a_3^2,$$

which is exactly the first Somos-4 step. The alternating recurrence (12) continues this mechanism.

The initial coefficients in (10) do more than reproduce the ordinary Hankel determinants. They also implement the companion prescription (4): the second product sequence attached to the Stieltjes fraction is fixed to be the same Somos-4 orbit with its first entry removed. The ordinary Hankel transform determines only the products $c_{2j-1}c_{2j}$ in (6); the companion sequence fixes the corresponding splitting into the odd and even Stieltjes coefficients. Consequently, once both product sequences and the normalization $c_0 = \lambda$ are prescribed, the Stieltjes fraction, and hence the selected moment sequence, is fully determined on the nondegenerate locus. This is the particular representative used throughout the paper.

A literal shifted determinant of the moments should nevertheless not be confused with the normalized companion product sequence just described. If

$$H_n^{(1)} = \det(g_{i+j+1})_{0 \leq i, j \leq n-1}, \quad (13)$$

then already $H_1^{(1)} = g_1 = \lambda c_1$, rather than c_1 unless $\lambda = 1$. Thus the phrase “the same orbit with its first entry removed” refers here to the normalized companion sequence fixed by the construction, not, without normalization, to the literal determinant (13).

The next step, is to eliminate the recursively defined continued-fraction tail and thereby obtain the explicit quadratic algebraic equation for G .

4 The algebraic generating function

Define the polynomials $A(z), B(z), C(z)$ by

$$A(z) = \lambda \eta m r^2 (\tau - \eta) z^3 \quad (14)$$

$$+ r \left(-(\lambda^2 \eta m + m^2(m + r^2))\tau + (2m - 1)\lambda^2 \eta^2 + \eta m^2(m + r^2) - \lambda r(m + r^2) \right) z^2 \quad (15)$$

$$+ \lambda \left(m^2 r^2 \tau - (m - 1)\lambda^2 \eta^2 - m(m^2 + 2mr^2 - m - r^2)\eta + \lambda r^3 \right) z \quad (16)$$

$$+ \lambda^2 \eta m(m - 1)r, \quad (17)$$

$$B(z) = -\lambda^2 \eta m r^2 \tau z^3 \quad (18)$$

$$+ r \left((2\lambda^3 \eta m + \lambda m^2(2m + r^2))\tau - (2m - 1)\lambda^3 \eta^2 - \lambda \eta m^3 + \lambda^2 r(2m + r^2) \right) z^2 \quad (19)$$

$$+ \lambda^2 \left(-2m^2 r^2 \tau + 2(m - 1)\lambda^2 \eta^2 + m(2m^2 + 2mr^2 - 2m - r^2)\eta - 2\lambda r^3 \right) z \quad (20)$$

$$- 2\lambda^3 \eta m(m - 1)r, \quad (21)$$

$$C(z) = -\lambda^2 m r \left(\tau(\lambda^2 \eta + m^2) + \lambda r \right) z^2 \quad (22)$$

$$+ \left(\lambda^3 m^2 r^2 \tau + \lambda^4 r^3 + \eta(m - 1)(\eta(-15\lambda^4 + 85\lambda^3 - 225\lambda^2 + 274\lambda - 120) \right. \quad (23)$$

$$\left. - \lambda^3 m^2 - 3(\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4)(\lambda - 5) \right) z \quad (24)$$

$$+ \lambda^4 \eta m(m - 1)r. \quad (25)$$

The generating function is the formal-power-series branch of

$$A(z)G(z)^2 + B(z)G(z) + C(z) = 0 \quad (26)$$

selected by the initial conditions implicit in (2), beginning with $g_0 = \lambda$ and $g_1 = m$.

Remark 1 (The apparently mysterious integer coefficients). The polynomial occurring in (25) is simply

$$-15\lambda^4 + 85\lambda^3 - 225\lambda^2 + 274\lambda - 120 = \frac{6!}{\lambda} \binom{\lambda}{6} - \lambda^5 \quad (27)$$

$$= (\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4)(\lambda - 5) - \lambda^5. \quad (28)$$

Thus the coefficients $-15, 85, -225, 274, -120$ arise by removing the leading term λ^5 from the falling factorial $(\lambda - 1) \cdots (\lambda - 5)$.

5 The two algebraic branches

With

$$D(z) = B(z)^2 - 4A(z)C(z),$$

the two algebraic branches may be represented by

$$G(z) = \frac{-B(z) \pm \sqrt{D(z)}}{2A(z)}. \quad (29)$$

When the constant term of A vanishes, the rationalized expressions

$$\frac{-2C(z)}{B(z) \mp \sqrt{D(z)}}$$

can be used instead. The sign in (29) cannot be fixed uniformly over the full parameter space.

Write

$$P(z, X) = A(z)X^2 + B(z)X + C(z).$$

The constant coefficients of (17)–(25) are

$$A_0 = \lambda^2 \eta m(m-1)r, \quad B_0 = -2\lambda^3 \eta m(m-1)r, \quad C_0 = \lambda^4 \eta m(m-1)r.$$

Consequently,

$$P(0, X) = A_0(X - \lambda)^2. \quad (30)$$

Both algebraic branches therefore coalesce at $X = \lambda$ when $z = 0$, and

$$\partial_X P(0, \lambda) = 2A_0\lambda + B_0 = 0. \quad (31)$$

The simple-root form of the formal implicit-function theorem is not applicable at the origin. The desired branch is instead selected by expanding the square root and requiring the prescribed coefficients, beginning with $g_0 = \lambda$ and $g_1 = m$.

Set

$$G(z) = \lambda + zF(z), \quad F(z) = \sum_{q \geq 0} f_q z^q, \quad f_q = g_{q+1}. \quad (32)$$

For

$$D_j = 2\lambda A_j + B_j, \quad E_j = \lambda^2 A_j + \lambda B_j + C_j,$$

where $C_3 = 0$, substitution into $P(z, G(z)) = 0$ gives

$$P(z, \lambda + zF) = \sum_{j=0}^3 E_j z^j + z \left(\sum_{j=0}^3 D_j z^j \right) F + z^2 \left(\sum_{j=0}^3 A_j z^j \right) F^2.$$

Equation (30) gives $E_0 = D_0 = 0$. An ordinary power-series branch additionally requires

$$E_1 = 0. \quad (33)$$

The next coefficient equation is

$$A_0 f_0^2 + D_1 f_0 + E_2 = 0. \quad (34)$$

Since $f_0 = g_1 = m$, the prescribed value of m selects one of the two branches of the desingularized quadratic.

6 Hankel-level shift of the parameters

To display the shift independently of the letters used above, let

$$G_{0;\ell,m,r,n,t}(x) = \sum_{q \geq 0} g_q x^q$$

denote the selected formal-power-series solution of (26) after the substitutions

$$\lambda = \ell, \quad \eta = n, \quad \tau = t.$$

Its Hankel determinants satisfy

$$(H_1, H_2, H_3, H_4) = (\ell, m, r, n), \quad H_{q+4}H_q = H_{q+3}H_{q+1} + tH_{q+2}^2. \quad (35)$$

The first Somos step gives

$$H_5 = \frac{H_2 H_4 + t H_3^2}{H_1} = \frac{mn + tr^2}{\ell}. \quad (36)$$

Hence shifting the Hankel sequence induces the parameter map

$$(\ell, m, r, n, t) \mapsto \left(m, r, n, \frac{mn + tr^2}{\ell}, t \right). \quad (37)$$

This concerns the Hankel transforms only. In particular, it does not imply an identity of generating functions such as

$$\frac{G_{0;\ell,m,r,n,t}(x) - g_0}{x} = G_{0;m,r,n,(mn+tr^2)/\ell,t}(x).$$

Distinct moment sequences, and hence distinct generating functions, may have the same Hankel transform.

7 A recurrence for the moments

Using the notation of (32), division of the expanded quadratic equation by z^2 yields

$$A(z)F(z)^2 + (D_1 + D_2 z + D_3 z^2)F(z) + E_2 + E_3 z = 0. \quad (38)$$

The branch condition is

$$A_0 m^2 + D_1 m + E_2 = 0, \quad (39)$$

and the branch is simple precisely when

$$2A_0 m + D_1 \neq 0. \quad (40)$$

Equivalently,

$$(2A_0 m + D_1)^2 = D_1^2 - 4A_0 E_2. \quad (41)$$

Define

$$T_q = \sum_{k=0}^q f_k f_{q-k}, \quad T_q = 0 \quad (q < 0). \quad (42)$$

Coefficient extraction from (38) gives

$$\sum_{j=0}^3 A_j T_{q-j} + D_1 f_q + D_2 f_{q-1} + D_3 f_{q-2} + E_{q+2} = 0, \quad (43)$$

where $f_q = 0$ for $q < 0$ and $E_j = 0$ for $j > 3$. For $q \geq 1$,

$$T_q = 2m f_q + \sum_{k=1}^{q-1} f_k f_{q-k}. \quad (44)$$

Under condition (40), the moments are therefore generated recursively by

$$f_q = - \frac{A_0 \sum_{k=1}^{q-1} f_k f_{q-k} + \sum_{j=1}^3 A_j T_{q-j} + D_2 f_{q-1} + D_3 f_{q-2} + E_{q+2}}{2A_0 m + D_1}. \quad (45)$$

This quadratic convolution recurrence follows directly from the algebraic generating-function equation. On the exceptional locus $2A_0 m + D_1 = 0$, the two desingularized branches coalesce at the origin and the next moment is not determined by the first-order coefficient equation; higher-order compatibility or a Puiseux expansion may then be required.

Because F is algebraic of degree at most two, it is D-finite. Consequently, the same moment sequence also satisfies a fixed-width linear recurrence with polynomial coefficients in the index.

8 Nonuniqueness of the moment representation

The ordinary Hankel transform does not in general determine the moment sequence. If only (H_n) is prescribed, the factorization $c_{2j-1}c_{2j}$ visible in (6) leaves freedom in the individual Stieltjes coefficients. The five-parameter algebraic generating function should therefore not be interpreted as the unique moment representation determined by the ordinary Somos–4 Hankel orbit alone.

The construction in this paper removes precisely this freedom by imposing the additional companion condition (4). In other words, we prescribe two product sequences: the first is the Somos–4 sequence (H_n) , and the second is the same sequence with its first value removed. Together with $c_0 = \lambda$, these two prescriptions determine the odd and even Stieltjes coefficients separately through (10) and (12). Hence the algebraic generating function presented in Section 3 is the fully specified representative selected by those two Hankel-level requirements. This choice was already built into the initial Stieltjes coefficients in Section 2; stating it explicitly here makes clear how the apparent degree of freedom is eliminated.

Other moment sequences can still share the first, ordinary Hankel transform when the companion condition is not imposed. For example, the binomial transform, a standard Hankel-transform-preserving operation[14, 7],

$$g'_q = \sum_{k=0}^q \binom{q}{k} g_k \quad (46)$$

has generating function

$$G'(z) = \frac{1}{1-z} G\left(\frac{z}{1-z}\right)$$

and preserves the Hankel determinants. Scalar and exponential rescalings of moments produce explicit normalization factors in the determinants, while affine changes of the underlying variable give further related moment functionals. Equality of Hankel transforms must therefore not be replaced by equality of generating functions without an independent argument.

No classification of the moment representations obtained after dropping the companion condition is attempted here. The inverse problem of describing all moment sequences realizing a given ordinary Hankel sequence remains open.

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